

Mutual Fund Analytics Platform

Bluestock Internship Capstone Project

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EXECUTIVE SUMMARY

The Mutual Fund Analytics Platform was developed as part of the Bluestock Internship Capstone Project to provide a comprehensive analytical framework for evaluating mutual fund performance, investor behavior, industry growth, and portfolio risk. The project addresses the growing need for data-driven decision-making in the Indian mutual fund industry, which has witnessed significant expansion due to increasing retail participation and systematic investment plan (SIP) adoption.

The primary objective of this project was to design and implement an end-to-end analytics solution capable of transforming raw financial data into meaningful business insights. The solution incorporates data engineering, exploratory data analysis, financial performance measurement, risk analytics, investor segmentation, and business intelligence dashboarding. The project utilizes multiple datasets including fund information, NAV history, assets under management (AUM), SIP inflows, investor transactions, portfolio holdings, and category-wise fund flows.

An Extract, Transform, and Load (ETL) pipeline was developed to clean, validate, and standardize the data before analysis. Exploratory Data Analysis (EDA) was performed to understand industry trends, fund growth patterns, investor demographics, and portfolio structures. Advanced financial metrics such as Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR) were calculated to evaluate mutual fund performance and risk characteristics.

In addition to performance analysis, investor-focused analytics such as cohort analysis, SIP continuity analysis, and a rule-based fund recommendation system were implemented. These features provide valuable insights into investor behavior, retention patterns, and fund suitability based on risk appetite. A Power BI dashboard was developed to visualize key metrics and support interactive exploration of the data.

The completed solution demonstrates the practical application of data analytics, financial modeling, and business intelligence technologies in the mutual fund domain. The project highlights how analytical tools can assist investors, fund managers, and financial institutions in making informed investment decisions while improving transparency and operational efficiency.

INTRODUCTION

Mutual funds have emerged as one of the most important investment vehicles in modern financial markets. They enable individuals to invest in diversified portfolios managed by professional fund managers, thereby reducing risk and improving accessibility to capital markets. In India, the mutual fund industry has experienced remarkable growth over the past decade, driven by increased financial awareness, digital investment platforms, and systematic investment plans (SIPs).

The rapid expansion of the industry has resulted in the generation of massive volumes of financial and transactional data. Every day, mutual funds generate Net Asset Value (NAV) records, portfolio holdings information, transaction histories, assets under management statistics, and investor activity logs. While this data contains valuable insights, extracting meaningful information requires advanced analytical techniques and visualization tools.

Data analytics plays a critical role in transforming raw financial data into actionable insights. Through analytics, investors can evaluate fund performance, compare investment alternatives, assess risk exposure, and identify long-term trends. Fund management companies can leverage analytics to improve portfolio allocation strategies, monitor investor behavior, and optimize product offerings.

One of the key drivers of mutual fund growth in India is the Systematic Investment Plan (SIP). SIPs allow investors to contribute fixed amounts periodically, promoting disciplined investing and wealth creation. The increasing popularity of SIPs has significantly contributed to industry growth and expanded participation among retail investors.

This project aims to build a comprehensive Mutual Fund Analytics Platform capable of analyzing multiple aspects of the mutual fund ecosystem. The platform integrates data engineering, statistical analysis, risk assessment, and dashboard development to create a unified analytical solution. The project demonstrates how modern data science techniques can be applied in financial services to enhance decision-making and improve investment outcomes.

PROJECT OBJECTIVES

The central goal of this initiative is the creation of an advanced, data-centric analytical ecosystem designed to deliver deep perspectives on the performance of mutual funds, the patterns of investor engagement, and broader market transformations.

Key technical and business goals include:

1. Aggregating and synthesizing diverse financial data streams into a cohesive environment for analysis.
2. Architecting a high-performance ETL framework for rigorous data cleansing, structural transformation, and integrity checks.
3. Conducting in-depth exploratory analysis to uncover critical industry trajectories and growth correlations.
4. Quantifying essential success indicators such as CAGR, alongside risk-adjusted metrics like the Sharpe and Sortino Ratios, Alpha, Beta, and downside volatility.
5. Evaluating tail-risk exposure through sophisticated modeling techniques, specifically Value at Risk and its conditional variants.
6. Examining the intersection of investor profiles, historical transaction flows, and the persistence of systematic investment strategies.
7. Constructing a logic-driven advisory engine that matches asset selection with individual risk thresholds.
8. Building dynamic Power BI interfaces to facilitate business intelligence and enhance strategic oversight.
9. Deriving meaningful, data-backed conclusions that empower sophisticated investment choices.
10. Exhibiting the functional utility of advanced analytics within the broader landscape of financial services.

DATA SOURCES

The project leverages a multifaceted array of data repositories, each offering a distinct perspective on the various layers of the investment fund sector.

Fund Master Dataset

Acting as the foundational catalog for all scheme-related inquiries, this repository includes granular details on fund houses, categorization, historical returns, efficiency ratios, and qualitative rankings.

NAV History Dataset

Spanning the period from 2022 to 2026, this log of daily valuation records provides the necessary time-series data to model periodic returns and evaluate risk-adjusted performance over varying horizons.

Assets Under Management Dataset

By monitoring the progression of managed capital across different entities, this dataset allows for an assessment of market dominance and the velocity of institutional expansion.

SIP Inflow Dataset

This archive records the volume of recurring contributions and active account counts, offering essential clues regarding the consistency of retail investment and general market sentiment.

Investor Transactions Dataset

Comprising anonymized behavioral data, this source tracks demographic variables, income brackets, and geographic trends to facilitate sophisticated segmentation and behavioral modeling.

Portfolio Holdings Dataset

These records disclose specific equity and debt allocations, permitting a detailed review of sectoral focus, asset concentration, and internal diversification strategies.

The synthesis of these diverse datasets ensures a holistic understanding of fund performance, client engagement, and overarching competitive forces.

ETL PIPELINE DESIGN

The Extract, Transform, and Load (ETL) architecture serves as the fundamental engine driving the Mutual Fund Analytics Platform. Given the aggregation of heterogeneous datasets from disparate origins, the establishment of a rigorous data preparation protocol was essential. This pipeline ensures the longitudinal consistency, reliability, and analytical readiness of the information used across all subsequent modeling phases.

Data Extraction

The initial phase involved the systematic retrieval of data from diverse CSV repositories representing the varied dimensions of the investment ecosystem. These inputs encompassed master scheme records, historical NAV logs, managed asset statistics, recurring contribution flows, and granular portfolio allocation details.

Leveraging the computational efficiency of Python's Pandas library, each data stream was ingested and rigorously inspected to map its underlying schema and structural characteristics prior to transformation.

Data Validation

Preceding the transformation stage, comprehensive integrity audits were conducted to isolate null values, redundant entries, and formatting anomalies. Such validation protocols are critical to safeguarding the accuracy of final analytical outputs against the risks of substandard data quality.

Primary validation metrics included:

- Rigorous auditing of mandatory data fields
- Automated detection of record duplication
- Standardization of temporal data formats
- Identification of outlier or unrealistic valuations
- Structural schema verification

Data Cleaning

The cleaning stage focused on the remediation of structural inconsistencies and the elimination of erroneous records within the datasets.

Core procedural refinements included:

- Systematic purging of redundant entries
- Unified normalization of column identifiers

- Strategic imputation of missing information
- Alignment of disparate formatting styles
- Precise casting of data types
- Filtering of non-compliant data points

This rigorous purification process enhanced the overall integrity of the information, establishing a robust foundation for sophisticated processing.

Data Transformation

Transformation logic converted raw inputs into synthesized, analysis-ready structures. Notable operations included:

- Mapping temporal strings to standardized datetime objects
- Deriving periodic returns from historical NAV sequences
- Synthesizing CAGR and various risk-adjusted metrics
- Harmonizing fund classification categories
- Assigning behavioral cohort labels to investor profiles
- Computing dynamic rolling performance indicators

These structural shifts enabled the application of advanced statistical modeling and interactive visual discovery.

Data Loading

The terminal phase involved persisting the refined data into a structured production repository. This curated environment serves as the authoritative source for exploratory analysis, financial simulations, and strategic dashboarding.

The design of the ETL pipeline guarantees the reproducibility, scalability, and long-term maintainability of the project's analytical framework.

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was executed to characterize data distributions, map emerging trajectories, and isolate underlying behavioral patterns. This phase established the empirical basis for developing the business intelligence dashboards and predictive models.

NAV Trend Analysis

The Net Asset Value (NAV) serves as the primary metric for unit-level fund valuation. Temporal sequences from 2022 through 2026 were examined to quantify sustained growth trajectories.

Findings indicated that equity-focused schemes demonstrated substantial capital appreciation during bullish cycles. While periodic volatility mirrored broader market sentiment, the long-term trend highlighted significant resilience and value creation.

Visualizing these valuation trends facilitated the identification of performance stability across diverse investment vehicles.

Assets Under Management (AUM) Analysis

Managed capital volumes represent the total market capitalization supervised by fund entities, serving as a proxy for market sentiment and organizational scale.

The analysis documented an uninterrupted expansion of AUM throughout the observed timeframe. While dominant fund houses maintained significant market share, smaller entities displayed consistent growth rates.

The progression of AUM underscores the deepening penetration of mutual funds and increasing retail participation within the financial ecosystem.

SIP Inflow Analysis

Systematic Investment Plans (SIPs) have evolved into the cornerstone of retail investment growth within the Indian landscape.

Monthly contribution logs revealed a persistent upward trend in recurring investor commitments. Notably, inflows reached a historic milestone of ₹31,002 Crore in December 2025, signaling high levels of retail persistence and disciplined capital allocation.

This momentum in SIP activity reflects a maturing investor base focused on long-term wealth accumulation strategies.

Category Inflow Analysis

Investor appetite fluctuates across various asset classes in response to shifting economic conditions and market cycles.

Granular analysis demonstrated that equity-oriented categories secured the most significant capital flows, particularly during expansionary phases. Conversely, debt and hybrid segments exhibited more moderated but stable inflow patterns.

The use of heatmaps provided critical insights into shifting category preferences and seasonal allocation behaviors.

Investor Demographics Analysis

This dimension of the analysis explored the intersection of age distribution, gender representation, income tiers, and regional factors.

Data revealed that the active workforce constitutes the primary investor cohort. Moderate-income segments showed strong adherence to SIP models, while higher-income individuals were characterized by larger discrete transaction volumes.

These demographic perspectives offer essential intelligence for refined market segmentation and targeted product development.

Geographic Analysis

Spatial analysis investigated the distribution of mutual fund engagement across regional borders and urban classifications.

While metropolitan hubs remain the primary drivers of investment volume, Tier-II and Tier-III cities are displaying accelerated participation. This shift underscores the success of digital platforms in democratizing financial access across the country.

Folio Growth Analysis

A folio represents the unique account identifier maintained for each investor by the fund management entity.

Aggregate folio counts surged from approximately 13.26 Crore in early 2022 to over 26.12 Crore by the end of 2025. This doubling of the account base reflects a transformative era of retail inclusion in professional investment vehicles.

Correlation Analysis

Statistical correlation matrices were derived using daily return sequences from representative scheme clusters.

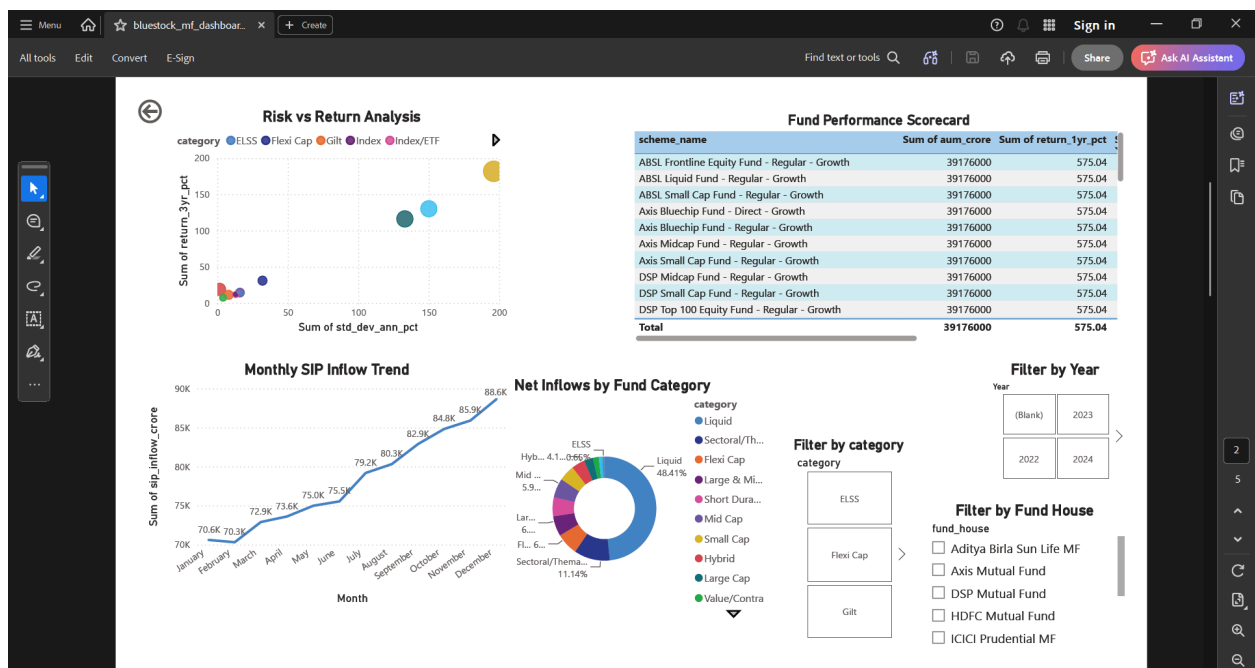
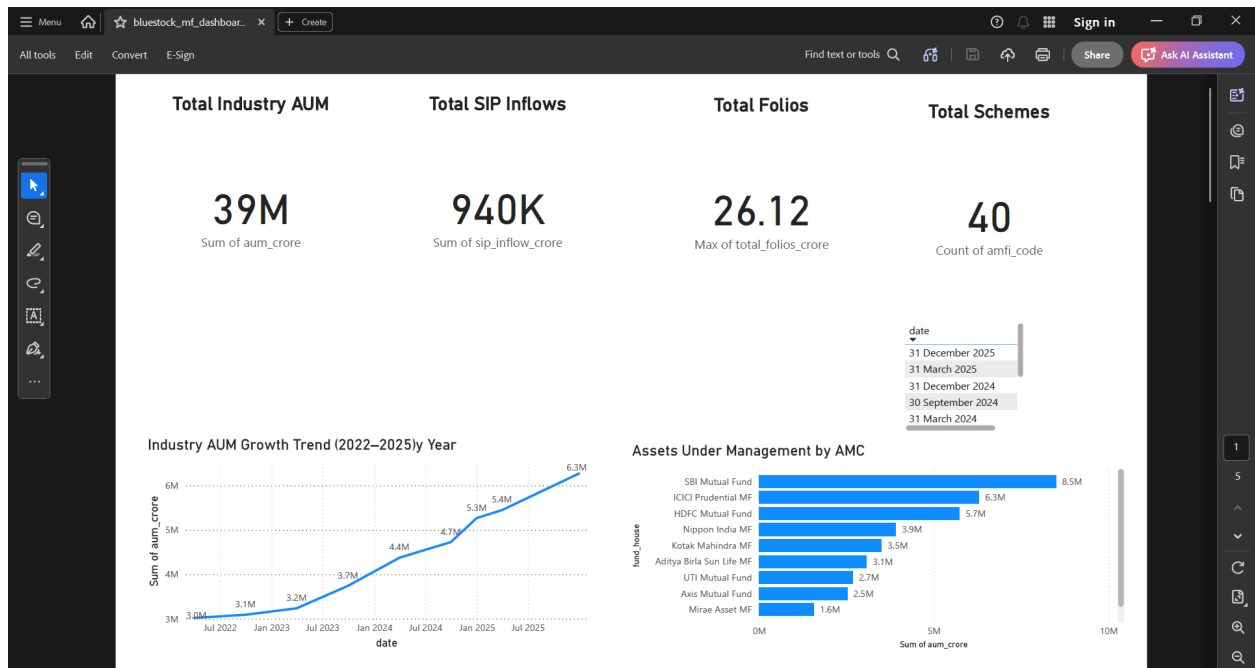
Results indicated strong positive relationships within intra-category funds, while cross-category pairings often displayed lower coefficients. This confirms the functional necessity of asset diversification in optimizing portfolio risk profiles.

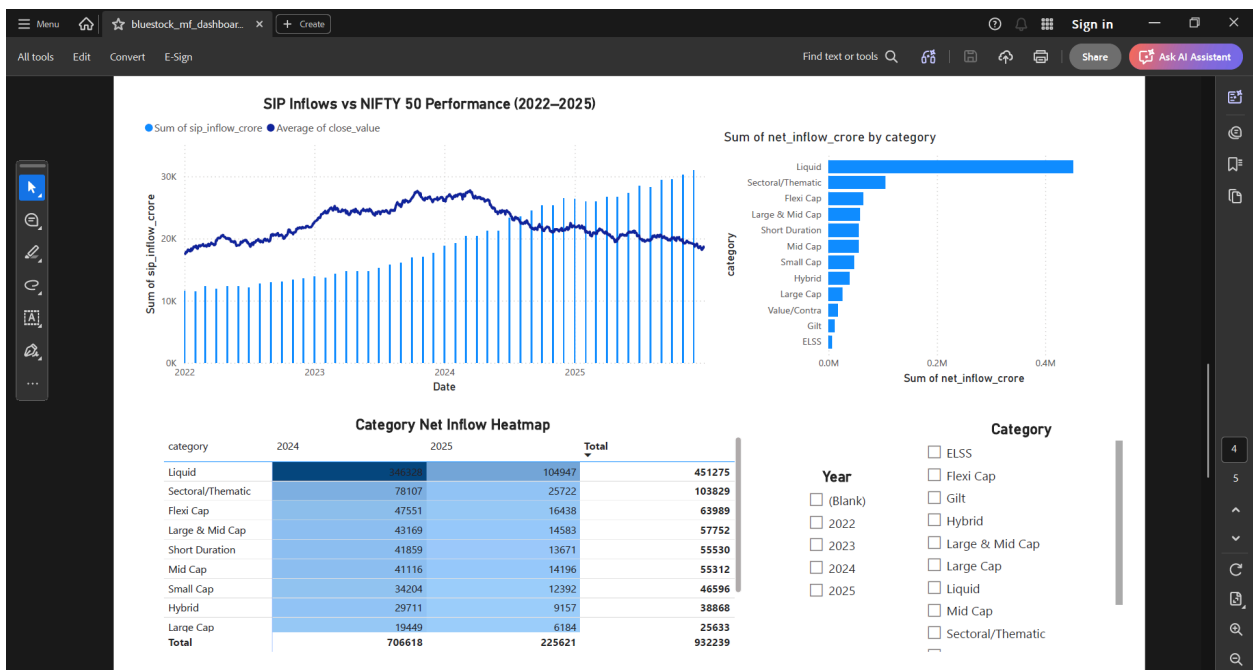
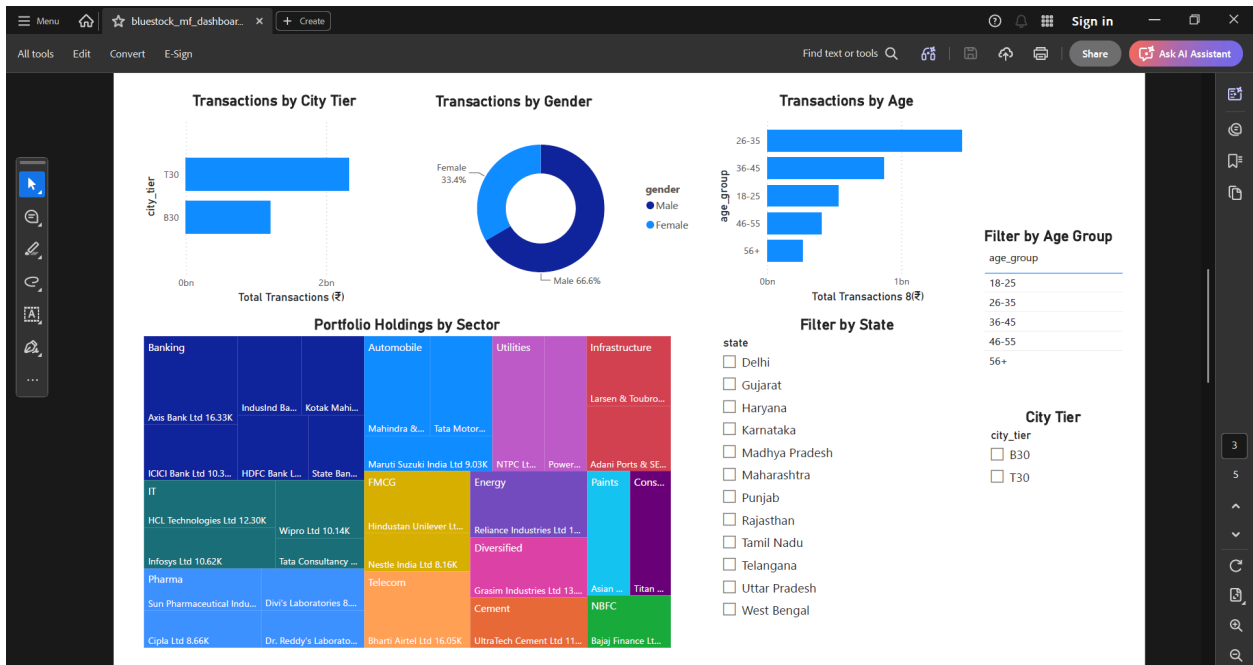
Sector Allocation Analysis

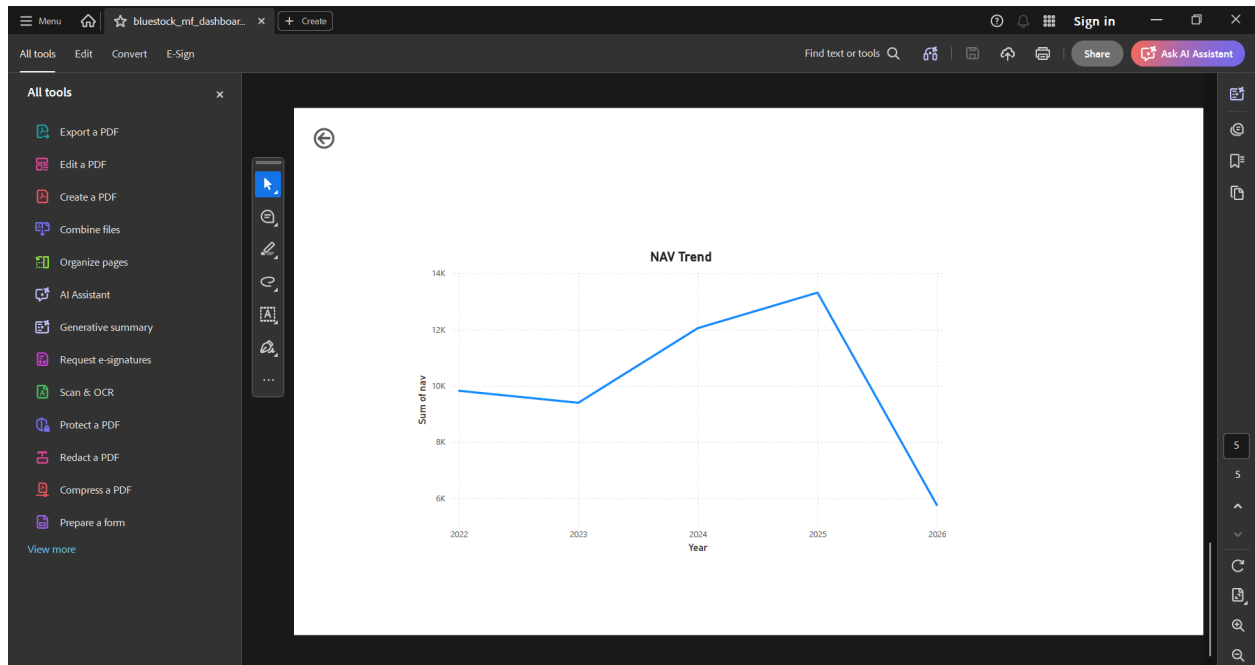
Portfolio holding snapshots were synthesized to map the sectoral orientation across the mutual fund spectrum.

Major exposures were identified in the Technology, Financial Services, Healthcare, and Industrial sectors. Differences in sectoral concentration highlight the varied management styles and risk tolerances inherent in different fund strategies.

This review also clarified patterns of asset concentration and internal diversification strategies.







KEY EDA FINDINGS

1. Fund valuations demonstrated consistent long-term appreciation despite intermittent volatility cycles.
2. Managed capital volumes displayed a sustained expansion, indicative of strengthening market confidence.
3. Record-breaking systematic inflows underscore the deepening commitment of retail investors.
4. Equity-based categories remain the primary attractors of new capital commitments.
5. Engagement patterns expanded significantly across diverse socioeconomic and age cohorts.
6. Metropolitan areas dominate volumes, while emerging urban centers show high growth velocity.
7. The industry account base experienced an unprecedented near-doubling during the study period.
8. Empirical correlation data validates the strategic efficacy of asset category diversification.
9. Specific economic sectors remain central to the allocation strategies of leading equity funds.
10. Market corrections proved to be transient, failing to disrupt the overarching growth trend.

PERFORMANCE ANALYTICS

Performance analytics was conducted to evaluate the effectiveness, stability, and risk-adjusted returns of mutual fund schemes. While absolute returns provide an indication of profitability, investors must also consider the risks undertaken to generate those returns. Therefore, multiple performance metrics were calculated to provide a comprehensive evaluation of mutual fund performance.

Daily Return Analysis

Daily returns were calculated using Net Asset Value (NAV) data. Daily returns represent the percentage change in NAV from one trading day to the next.

The analysis of daily returns revealed that most funds exhibited average daily returns close to zero with periodic fluctuations influenced by market conditions. While daily movements were relatively small, cumulative effects over long investment horizons resulted in substantial wealth creation.

Daily return distributions also provided valuable insights into fund volatility and downside risk.

Compound Annual Growth Rate (CAGR)

Compound Annual Growth Rate (CAGR) measures the average annual growth rate of an investment over a specified period while accounting for compounding effects.

CAGR was calculated for all mutual fund schemes using historical NAV data. The results showed that equity-oriented schemes generally delivered higher long-term returns compared to debt-oriented funds. Small-cap and mid-cap funds demonstrated the highest CAGR values, reflecting their growth-oriented investment strategies.

The CAGR analysis helped identify funds capable of generating consistent long-term wealth creation for investors.

Sharpe Ratio Analysis

The Sharpe Ratio is one of the most widely used measures of risk-adjusted performance. It evaluates how much excess return a fund generates for each unit of risk undertaken.

A risk-free rate of 6.5% was used as a proxy based on prevailing RBI policy rates.

Funds with higher Sharpe Ratios demonstrated superior efficiency in converting risk into returns. Several large-cap and diversified equity funds achieved Sharpe Ratios greater than one, indicating strong risk-adjusted performance.

The analysis revealed that high returns alone do not necessarily imply superior performance. Some funds achieved comparable returns while exposing investors to significantly lower levels of risk.

Sortino Ratio Analysis

While the Sharpe Ratio considers total volatility, the Sortino Ratio focuses specifically on downside volatility. Since investors are generally more concerned about losses than gains, the Sortino Ratio provides a more targeted measure of downside risk.

Funds with high Sortino Ratios demonstrated strong downside protection while maintaining attractive returns. The results indicated that several funds achieved significantly higher Sortino Ratios than Sharpe Ratios, suggesting that their volatility was primarily driven by positive returns rather than losses.

The Sortino analysis provided additional confidence in identifying high-quality investment opportunities.

Alpha Analysis

Alpha measures a fund manager's ability to generate returns beyond those expected based on market movements.

Positive Alpha indicates outperformance relative to the benchmark, while negative Alpha suggests underperformance.

The analysis identified several funds with consistently positive Alpha values, indicating effective portfolio management and stock selection capabilities. These funds generated returns above what would normally be expected based on their market exposure.

Funds with higher Alpha values may provide greater value to investors through active management strategies.

Beta Analysis

Beta measures the sensitivity of a fund to overall market movements.

A Beta greater than one indicates that the fund tends to move more aggressively than the market, while a Beta less than one suggests lower sensitivity.

The Beta analysis revealed that small-cap and aggressive equity funds generally exhibited higher market sensitivity. Conversely, debt-oriented and conservative funds demonstrated lower Beta values.

Understanding Beta helps investors align fund selection with their risk tolerance and market outlook.

Maximum Drawdown Analysis

Maximum Drawdown measures the largest peak-to-trough decline experienced by a fund during the analysis period.

This metric provides a realistic assessment of downside risk by showing the worst historical loss investors could have experienced.

The analysis identified significant differences in drawdown behavior across categories. Small-cap funds experienced larger drawdowns due to higher volatility, while diversified and debt-oriented funds demonstrated relatively lower declines.

Maximum Drawdown is particularly important for investors seeking capital preservation and downside protection.

Fund Scorecard Development

To create a unified ranking framework, a Fund Scorecard was developed using multiple performance indicators.

The scoring model incorporated:

- 30% Weight – Three-Year Return Ranking
- 25% Weight – Sharpe Ratio Ranking
- 20% Weight – Alpha Ranking
- 15% Weight – Expense Ratio Ranking (Inverse)
- 10% Weight – Maximum Drawdown Ranking (Inverse)

This composite approach ensured that fund rankings reflected both return generation and risk management capabilities.

The scorecard enabled investors to compare funds using a single performance metric while considering multiple dimensions of performance.

Benchmark Comparison

Top-performing funds were compared against market benchmarks to evaluate relative performance.

The comparison demonstrated that several actively managed funds outperformed benchmark indices over extended periods. Tracking error analysis further quantified deviations between fund returns and benchmark returns.

These findings provided valuable insights into active management effectiveness and investment strategy execution.

Performance Analysis Findings

1. Small-cap funds generated the highest long-term returns among analyzed schemes.
2. Several large-cap funds achieved superior risk-adjusted performance with Sharpe Ratios above one.
3. Positive Alpha values indicated successful active management in multiple funds.
4. Higher-return funds generally exhibited higher volatility and larger drawdowns.
5. Diversified portfolios demonstrated stronger downside protection.
6. Expense ratios influenced long-term investor returns and scorecard rankings.
7. Benchmark comparisons revealed that selected funds consistently outperformed market indices.
8. Risk-adjusted performance metrics provided more reliable fund rankings than return-based measures alone.
9. Maximum Drawdown analysis highlighted significant differences in downside risk across categories.
10. Composite scorecards provided a balanced framework for mutual fund evaluation.

ADVANCED ANALYTICS

While traditional performance measures provide valuable information, advanced analytics offer deeper insights into risk, investor behavior, and portfolio characteristics. This project incorporated several advanced analytical techniques to enhance decision-making capabilities.

Historical Value at Risk (VaR)

Value at Risk (VaR) estimates the maximum expected loss over a specified time period at a given confidence level.

A 95% confidence interval was used in this project to estimate potential daily losses for each mutual fund.

The analysis identified several high-risk funds with VaR values exceeding -2.5%, indicating greater exposure to adverse market conditions. These funds may generate attractive returns but also expose investors to higher downside risk.

VaR provides investors with a practical measure of potential loss under normal market conditions.

Conditional Value at Risk (CVaR)

Conditional Value at Risk extends VaR by estimating the average loss during extreme market events.

CVaR values were consistently larger than corresponding VaR values, highlighting the severity of losses during adverse market scenarios.

This analysis provided a more realistic assessment of tail risk and helped identify funds vulnerable to extreme market declines.

Rolling 90-Day Sharpe Ratio

Risk-adjusted performance can vary significantly over time. To capture these dynamics, rolling 90-day Sharpe Ratios were calculated for top-performing funds.

The analysis revealed periods of strong performance as well as temporary declines in risk-adjusted returns. Funds with consistently positive rolling Sharpe Ratios demonstrated greater stability and resilience across varying market conditions.

Rolling metrics provide investors with a dynamic perspective on fund performance rather than relying solely on historical averages.

Investor Cohort Analysis

This segment of the study focused on characterizing investor behavior by grouping individuals based on the year of their initial market entry. By categorizing participants into specific cohorts

according to their first transaction date, a longitudinal review of engagement patterns and evolving participation trajectories was facilitated.

The empirical evidence showed that the 2024 cohort represented the most significant portion of the investor base and was responsible for the largest share of capital commitments. This surge indicates a transformative era for retail inclusion and highlights the accelerating interest in professionally managed investment vehicles during the study period.

Furthermore, a comparative review of average capital allocations across these groups revealed that more recent cohorts exhibited higher transaction velocities and stronger activity levels. Such trends underscore a deepening sense of market confidence in diverse fund offerings.

The derived perspectives offer essential intelligence for asset management entities looking to refine their strategies for client acquisition and long-term retention.

SIP Continuity Analysis

Systematic Investment Plans (SIPs) serve as a primary proxy for evaluating the persistence and commitment of the retail sector. To quantify the consistency of these recurring contributions, a rigorous continuity audit was performed.

The evaluation focused on participants with a minimum of six systematic transactions. By modeling the temporal intervals between consecutive contributions, investors with an average hiatus exceeding thirty-five days were classified as "At Risk" of disengagement.

The analysis documented that a substantial portion of the base displayed irregular allocation behavior, suggesting potential attrition or a decline in active engagement. Only a limited segment of the population maintained perfectly synchronized investment cycles.

These observations underscore the functional necessity of proactive engagement frameworks, automated notification systems, and financial literacy programs to strengthen SIP persistence and loyalty.

Fund Recommendation Engine

A logic-driven advisory framework was engineered to match individuals with suitable mutual fund schemes based on their specific risk thresholds.

Asset selections were organized into distinct risk classifications:

- Low Risk
- Moderate Risk

- Moderately High Risk
- High Risk
- Very High Risk

Within each tier, schemes were prioritized using the Sharpe Ratio, ensuring that suggestions prioritized efficient risk-adjusted performance over simple historical profitability.

This advisory engine demonstrates the practical utility of analytics in delivering personalized investment guidance and enhancing the digital user experience within financial platforms.

Sector HHI Concentration Analysis

Effective risk management is fundamentally dependent on strategic portfolio diversification. To quantify the level of asset concentration, the Herfindahl-Hirschman Index (HHI) was applied to the available sectoral allocation data.

Funds characterized by elevated HHI coefficients displayed significant exposure to a narrow cluster of sectors, increasing their vulnerability to idiosyncratic market shocks. In contrast, schemes with lower index values exhibited more robust internal diversification strategies.

The review isolated Fund 119092 as possessing the highest concentration risk among the studied schemes. Investors focused on broad exposure may find greater utility in funds demonstrating lower HHI scores.

Utilizing HHI analysis provides a critical layer of risk oversight that complements traditional performance-based evaluations.

Advanced Analytics Findings

1. Various analyzed schemes showed substantial exposure to downside volatility according to VaR and CVaR modeling.
2. Tail-risk assessments confirmed that extreme market events could trigger losses far exceeding standard daily fluctuations.
3. Rolling risk-adjusted metrics highlighted the dynamic and shifting nature of fund performance over time.
4. Longitudinal cohort data demonstrated an accelerating trend in retail participation and commitment levels.
5. Continuity analysis identified a significant portion of the base that may require intervention to maintain systematic consistency.
6. Sophisticated recommendation systems can effectively align asset selection with individual risk profiles.

7. The degree of sectoral concentration varies widely across the mutual fund landscape.
8. Diversified fund structures typically exhibited lower levels of concentration-related risk.
9. Advanced modeling techniques uncovered perspectives that are often obscured in traditional return-based reviews.
10. The integration of behavioral and risk-focused analytics enhances the quality of strategic investment choices.